

Sym:

Items in this listing are coded by a symbol indicating the status of the item. These codes are:

- C Required item for FAA Certification.
- S Standard equipment. Most standard equipment is applicable to all airplanes. Some equipment may be replaced by optional equipment.
- O Optional equipment. Optional equipment may be installed in addition to or to replace standard equipment.

Qty: The quantity of the listed item in the airplane. A hyphen (-) in this column indicates that the equipment was not installed.

ATA Item	Description	SYM	QTY	Part Number	Unit Weight	Arm
34-08	GPS 1 Antenna	C	1	12744-001	0.4	136.2
34-09	GPS 2 Antenna	S	1	12744-001	0.4	110.3
34-10	Transponder Antenna	C	1	12739-001	0.1	105.0
34-11	VOR/LOC Antenna	C	1	12742-001	0.4	331.0
34-12	Turn coordinator, modified	C	1	11891-001	1.8	118.0
34-13	GMA 340 audio panel	S	1	12717-050	1.5	121.5
34-14	GNS 420 (GPS/COM/NAV)	O	1	12718-004	5.0	121.0
34-15	GNS 420 (GPS/COM/NAV)	C	1	12718-051	5.0	121.0
34-16	GNS 420 (GPS/COM/NAV)	O	1	12718-051	5.0	122.4
EMax engine monitoring						
34-17	• Data acquisition unit	O	1	16692-001	2.0	118.0
34-18	• Monitor cabin harness	O	1	16695-005	2.0	108.0
Sky watch option						
34-19	• Sky watch inverter	O	1	14484-001	0.5	118.0
34-20	• Sky watch antenna nsti	O	1	14480-001	2.3	150.5
34-21	• Sky watch track box	O	1	14477-050	10.0	140.0
Stormscope option						
34-22	• Processor	O	1	12745-050	1.7	199.0
34-23	• Antenna	O	1	12745-070	0.9	191.0
Transponder option						
34-24	• Mode A/C transponder	C	1	13587-001	1.6	124.9
34-25	• Mode S transponder	O	-	15966-050	2.6	121.0
TAWS option						
34-26	• KGP 560 processor	O	1	15963-001	1.3	117.0
XM satellite option						
34-27	• XM WX/radio receiver	O	1	16121-001	1.7	114.0
34-28	• XM radio remote control	O	1	16665-501	0.2	149.3
61	Propeller					
61-01	• Hartzell propeller installation	C	1	15319-00X	79.8	48.0
61-02	• McCauley propeller installation	O	1	15825-00X	78.0	50.0
61-03	• Propeller governor	C	1	15524-001	3.2	61.7
71	Power plant					
71-01	• Upper cowl	C	1	20181-003	10.5	78.4
71-02	• Lower cowl LH	C	1	20182-005	5.4	78.4
71-03	• Lower cowl RH	C	1	20439-005	5.4	78.4
71-03	• Engine baffling installation	C	1	15460-001	10.7	78.4

Figure 1-5. Example of some of the required standard or supplemental and optional equipment for an aircraft.

Flight Standards Service

The FAA's Flight Standards Service (FS) sets aviation standards for airmen and aircraft operations in the United States and for American airmen and aircraft around the world. Flight Standards is organized into four functional offices: Office of Safety Standards, Air Carrier Safety Assurance, General Aviation Safety Assurance, and Foundational Business.

The primary interface between FS and the general aviation community/general public is the local Flight Standards District Office (FSDO). The FSDOs are responsible for the certification and surveillance of certain air carriers, air operators, flight schools/training centers, airmen (pilots, flight instructors, mechanics and other certificate holders). FSDO inspectors also handle general aviation accident investigation at the request of, or in cooperation with, the National Transportation Safety Board.

Each FSDO is staffed by Aviation Safety Inspectors (ASIs) whose specialties include operations, maintenance, and avionics. General Aviation ASIs are highly qualified and experienced aviators. Once accepted for the position, an inspector will satisfactorily complete indoctrination training conducted at the FAA Academy. The indoctrination training coursework for a General Aviation Operations Inspector, which is oriented to the tasks to be performed by an ASI in the general aviation environment, includes classroom and flight training on pilot certification activities. Thereafter, the inspector will complete recurrent training on a regular basis. Among other duties, the ASI is responsible for administering FAA practical tests for pilot and flight instructor certificates and associated ratings. Questions concerning pilot certification and/or requests for other aviation information or services should be directed to the FSDO. For specific FSDO locations and telephone numbers, refer to www.faa.gov.

Role of the Pilot Examiner

Pilot and flight instructor certificates are issued by the FAA upon satisfactory completion of required knowledge and practical tests. The administration of practical tests is an FAA responsibility that may occur at the FSDO level. However, in order to satisfy the public need for pilot testing and certification services, the FAA delegates certain responsibilities, as the need arises, to private individuals who are not FAA employees. A Designated Pilot Examiner (DPE) is a private citizen who is designated as a representative of the FAA Administrator to perform specific (but limited) pilot certification tasks on behalf of the FAA and may charge a reasonable fee for doing so. Generally, a DPE's authority is limited to accepting applications and conducting practical tests leading to the issuance of specific pilot certificates and/or ratings. A DPE operates under the direct supervision of the FSDO that holds the examiner's designation file. A FSDO inspector is assigned to monitor the DPE's certification activities.

The FAA selects highly qualified individuals to be DPEs. These individuals have good industry reputations for professionalism, high integrity, a demonstrated willingness to serve the public, and adhere to FAA policies and procedures in certification matters. A DPE is expected to administer practical tests with the same degree of professionalism, using the same methods, procedures, and standards as an FAA ASI. Note that a DPE is not an FAA ASI. A DPE cannot initiate enforcement action, investigate accidents, or perform surveillance activities on behalf of the FAA. However, the majority of FAA practical tests at the recreational, private, and commercial pilot level are administered by DPEs.

Role of the Flight Instructor

The flight instructor is the cornerstone of aviation safety. The FAA has adopted an operational training concept that places the full responsibility for pilot training on the flight instructor. In this role, the instructor assumes the total responsibility for providing training in all the knowledge areas and skills necessary for pilots to operate safely and competently in the National Airspace System (NAS). This training includes airmanship skills, pilot judgment and decision-making, hazard identification, risk analysis, and good operating practices. (See Risk Management Handbook, FAA-H-8083-2). [Figure 1-6]



Figure 1-6. *The flight instructor is responsible for teaching and training.*

A flight instructor normally meets broad flying experience requirements, passes rigid knowledge and practical tests, and demonstrates the ability to apply recommended teaching techniques before being certificated.

A pilot training program is dependent on the quality of the ground and flight instruction given. A good flight instructor has a thorough understanding of the learning process, knowledge of the fundamentals of instruction, and the ability to communicate effectively with the learner.